

Advanced (Habanero)

- Q1.** What is the vertex of the graph of $y = |x - 2|$?
(A) (2, 0)
(B) (0, 2)
(C) (2, 2)
(D) (0, 0)
(E) (1, 1)
- Q2.** The graph of $y = |x + 1|$ has an axis of symmetry at $x =$
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q3.** The graph of $y = 2|x|$ opens
(A) upward
(B) downward
(C) to the left
(D) to the right
(E) not at all
- Q4.** The graph of $y = |x| + 2$ is a transformation of the graph of $y = |x|$ by a
(A) horizontal shift of 2 units
(B) vertical shift of 2 units
(C) horizontal stretch by a factor of 2
(D) vertical compression by a factor of 2
(E) vertical stretch by a factor of 2
- Q5.** The equation $y = |x - 3| - 2$ represents a graph that is
(A) the graph of $y = |x|$ shifted 3 units to the right and 2 units down
(B) the graph of $y = |x|$ shifted 3 units to the left and 2 units up
(C) the graph of $y = |x|$ shifted 3 units to the right and 2 units up
(D) the graph of $y = |x|$ shifted 3 units to the left and 2 units down
(E) the graph of $y = |x|$ shifted 2 units to the right and 3 units down
- Q6.** What is the equation of the graph that is the same as the graph of $y = |x|$ but shifted 2 units to the left and 1 unit down?
(A) $y = |x + 2| - 1$
(B) $y = |x - 2| + 1$
(C) $y = |x + 2| + 1$
(D) $y = |x - 2| - 1$
(E) $y = |x| + 2 - 1$
- Q7.** Which of the following is the equation of the graph of $y = |x|$ after it has been stretched vertically by a factor of 3?
(A) $y = 3|x|$
(B) $y = \frac{1}{3}|x|$
(C) $y = |3x|$
(D) $y = |x| + 3$
(E) $y = |x| - 3$
- Q8.** The equation $y = -|x|$ represents a graph that is
(A) the graph of $y = |x|$ reflected across the x-axis
(B) the graph of $y = |x|$ reflected across the y-axis
(C) the graph of $y = |x|$ shifted 1 unit to the right
(D) the graph of $y = |x|$ shifted 1 unit to the left
(E) the graph of $y = |x|$ shifted 1 unit up

Open Ended Questions (FRQ)

- Q9.** Graph the function $y = |x - 2| + 1$. Identify the vertex, axis of symmetry, and direction of opening.
- Q10.** Suppose the graph of $y = |x|$ is transformed into the graph of $y = a|x - h| + k$. Describe the effects of the parameters a , h , and k on the graph of $y = |x|$. Provide examples.

Chef's Tips

- Tip 1: Ensure students understand the properties of absolute value functions.
- Tip 2: Make sure students can apply transformations to absolute value graphs.
- Tip 3: Emphasize the importance of identifying vertex, axis of symmetry, and direction of opening.